Introduction to Probability & Statistics

A concise 6-page summary of core concepts and formulas

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1. Probability, Statistics & Information Theory

Probability measures the chance a random event occurs (a number in [0, 1]). **Statistics** analyzes data to make inferences about a population. **Information theory** quantifies the uncertainty/surprise of outcomes. A **statistical model** is a mathematical framework linking these ideas: it uses probability to describe data and is fit so that the *likelihood* of the observed data is maximized.

Random Variable (RV)

A variable whose value is determined by a random experiment. Maps outcomes → numbers.

- **Discrete RV**: countable values. Described by a **PMF** $p(x)$. $\Sigma\, p(x) = 1$.
- **Continuous RV**: any value in a range. Described by a **PDF** $f(x)$. $\int f(x)\, dx = 1$.
- **CDF**: $F(x) = P(X \leq x)$ - step for discrete, smooth for continuous.

Probability vs Likelihood

- **Probability** $P(X = x \mid \theta)$: chance of an outcome given fixed parameters θ.
- **Likelihood** $L(\theta \mid x)$: how well parameters θ explain observed data x.

A model generalizes well when θ is chosen so $L(\theta \mid x)$ is maximized (Maximum Likelihood Estimation).

Shannon Entropy - expected amount of information / surprise of an outcome drawn from distribution P(X):

$$H(X) = -\Sigma_i P(x_i) \log_2 P(x_i) \quad \text{[bits]}$$

Maximal for a uniform distribution (fair coin: H = 1 bit). Zero if the outcome is certain.

Concept	Focus	Typical use
Probability	Randomness, uncertainty	Predict event frequencies
Statistics	Data analysis, inference	Learn about a population from a sample
Stochastics	Time-dependent randomness	Markov chains, random processes
Likelihood	Parameter estimation	Fit models to observed data
Information theory	Quantifying information	Entropy, coding, compression

2. Descriptive Statistics

Organize, summarize, and visualize data. Start by distinguishing **population** (whole group, size N) from **sample** (subset, size n) and identifying the **type of data**:

Type	Subtype	Examples
Quantitative	Discrete (counts)	# students, # defective items
	Continuous (measurements)	Height, temperature, income
Qualitative	Nominal (no order)	Colors, gender, blood type
	Ordinal (ordered)	Survey rating, education level, rank

Measures of Central Tendency

Mean (arithmetic average): $\mu = (1/N)\ \Sigma_{i=1}^{N} x_i$ **Sample mean**: $\bar{x} = (1/n)\ \Sigma_{i=1}^{n} x_i$
Median: middle value of sorted data **Mode**: most frequent value
Expectation (discrete RV): $E[X] = \Sigma_i\ x_i\ P(X = x_i)$ **Continuous**: $E[X] = \int x\ f(x)\ dx$

Measures of Dispersion

Population variance: $\sigma^2 = (1/N)\ \Sigma_i\ (x_i - \mu)^2$ **Sample variance**: $s^2 = 1/(n-1)\ \Sigma_i\ (x_i - \bar{x})^2$
Standard deviation: $\sigma = \sqrt{(\sigma^2)}$ - spread around the mean, same units as the data.
IQR (interquartile range): $Q_3 - Q_1$ - robust to outliers.

Sample variance uses n−1 (Bessel's correction) for an unbiased estimate.

Shape of a Distribution

Skewness: asymmetry about the mean. Positive → long right tail; negative → long left tail.

Kurtosis: heaviness of tails vs the normal distribution. Outliers drive positive (excess) kurtosis.

$$\text{Skew} = E[((X - \mu) / \sigma)^3]$$
$$\text{Kurt} = E[((X - \mu) / \sigma)^4] - 3$$

Quantiles

Values that split ordered data into equal-probability groups:

- **Quartiles**: 4 parts (Q_1, Q_2=median, Q_3)
- **Deciles**: 10 parts **Percentiles**: 100 parts
- Summarize distributions and flag outliers (e.g., points beyond $Q_1 - 1.5 \cdot IQR$ or $Q_3 + 1.5 \cdot IQR$).

Visualization: histograms & density plots (distributions), bar & pie charts (categories), line charts (time series), scatter & bubble plots (relationships), box plots (quartiles + outliers), heat maps (matrices, geo-data, calendar views).

3. Combinatorics & Probability Distributions

Combinatorics - counting discrete structures

Combinatorics counts the number of ways to arrange or choose items. It underpins many probabilities in discrete settings (especially binomial and hypergeometric).

Permutation of n distinct items (order matters, no replacement): $n!$
k-permutation (order matters, choose k of n): $P(n, k) = n! / (n - k)!$
Combination (order does *not* matter, choose k of n): $C(n, k)$ = "n choose k" = $n! / [k! (n - k)!]$
Pascal's triangle: row n gives $C(n, 0), C(n, 1), \ldots, C(n, n)$; each entry is the sum of the two directly above.

Probability Distributions - PMF / PDF / CDF

A probability distribution describes how likely each outcome of a random variable is.

Aspect	Discrete	Continuous
Function	Probability Mass Function p(x)	Probability Density Function f(x)
Values	$p(x) = P(X = x)$ in [0, 1]	$f(x) \geq 0$ (density, not a probability)
Sum / integral	$\Sigma_x\, p(x) = 1$	$\int_{-\infty}^{\infty} f(x)\, dx = 1$
CDF	$F(x) = \Sigma_{t \leq x}\, p(t)$	$F(x) = \int_{-\infty}^{x} f(t)\, dt$
Interval prob.	$P(a \leq X \leq b) = \Sigma_{x=a}^{b}\, p(x)$	$P(a \leq X \leq b) = \int_a^b f(x)\, dx$
Visualization	Lollipop / bar chart	Smooth curve (e.g. bell curve)

Map of Common Distributions

Most classic distributions relate back to the Bernoulli trial. Knowing the family of your data guides both modeling and hypothesis tests.

Distribution	Type	Describes / relates to
Uniform	Discrete or continuous	Equal probability over an interval (fair die / coin)
Bernoulli	Discrete	One trial, success/failure
Binomial	Discrete	k successes in n independent Bernoulli trials
Hypergeometric	Discrete	Like binomial, but sampling without replacement
Poisson	Discrete	Events in fixed time/space, $n \to \infty$, rare events
Normal (Gaussian)	Continuous	Limit of Binomial ($n, k \to \infty$); Central Limit Theorem
Standard normal	Continuous	Normal with $\mu = 0, \sigma = 1$ (z-scores)
Student's t	Continuous	Like normal, heavier tails; small n, unknown σ
Exponential / Gamma	Continuous	Waiting time until 1st / k-th Poisson event

4. Key Distributions - Formulas

Bernoulli & Binomial

Bernoulli(p): $P(X = 1) = p$, $P(X = 0) = 1 - p$ $E[X] = p$, $Var(X) = p(1 - p)$
Binomial(n, p): $P(X = k) = C(n, k) \cdot p^k (1 - p)^{n-k}$ $k = 0, 1, \ldots, n$
$E[X] = n p$, $Var(X) = n p (1 - p)$

Example: a player with a 60% free-throw rate making exactly 7 of 10 shots has $P = C(10, 7)(0.6)^7(0.4)^3 \approx 0.215$.

Poisson

Poisson(λ): $P(X = k) = (\lambda^k e^{-\lambda}) / k!$ $k = 0, 1, 2, \ldots$
$E[X] = \lambda$, $Var(X) = \lambda$ (λ is the average rate)

Use when counting events in a fixed interval (calls/min, goals/match). Approximates the Binomial when n is large and p is small.

Normal (Gaussian) & Standard Normal

Normal(μ, σ²): $f(x) = (1 / (\sigma \sqrt{(2\pi)})) \cdot \exp(- (x - \mu)^2 / (2\sigma^2))$
z-score (standardization): $z = (x - \mu) / \sigma \rightarrow Z \sim N(0, 1)$
Empirical rule: ≈68% of values lie within $\mu \pm 1\sigma$, ≈95% within $\mu \pm 2\sigma$, ≈99.7% within $\mu \pm 3\sigma$.

Central Limit Theorem: sums and means of many independent RVs tend to be normal, regardless of the underlying distribution.

Student’s t-distribution

Used when the sample is small ($n < 30$) or the population standard deviation σ is unknown. Heavier tails than the normal; shape depends on **degrees of freedom** $df = n - 1$. As $df \to \infty$, the t-distribution approaches $N(0, 1)$.

Distribution	Parameters	Mean	Variance
Bernoulli	p	p	$p(1 - p)$
Binomial	n, p	$n p$	$n p (1 - p)$
Poisson	λ	λ	λ
Uniform [a, b]	a, b	$(a + b) / 2$	$(b - a)^2 / 12$
Normal	μ, σ^2	μ	σ^2
Exponential	λ	$1 / \lambda$	$1 / \lambda^2$

5. Hypothesis Testing & A/B Testing

Hypothesis testing uses sample data to decide whether there is enough evidence to reject a **null hypothesis H_0** ("no effect") in favor of an **alternative H_1**.

Standard workflow

1. Formulate H_0 and H_1 (one-sided or two-sided).
2. Choose a significance level α (commonly 0.05 or 0.01).
3. Pick the right test (see table below) and compute the test statistic.
4. Compute the **p-value** = P(observed result or more extreme | H_0).
5. Decision rule: if $p \le \alpha$ → reject H_0, else fail to reject.

One-sided vs two-sided tests

Two-sided: $H_0: \theta = \theta_0$ vs $H_1: \theta \neq \theta_0$ — rejection in both tails.
One-sided: $H_0: \theta \le \theta_0$ vs $H_1: \theta > \theta_0$ — rejection in one tail only (more power if direction is known).

Errors: false positives & false negatives

	H_0 is True	H_0 is False
Reject H_0	**Type I error** (false positive, prob. α)	**Correct** (power = 1 − β)
Fail to reject H_0	**Correct** (1 − α)	**Type II error** (false negative, prob. β)

Choosing the right test (A/B testing in practice)

Test	Use when...	Statistic / formula
Two-sample t-test (independent)	Compare means of two independent groups (continuous, ≈ normal).	$t = (\bar{x}_1 - \bar{x}_2) / \sqrt{s_1^2/n_1 + s_2^2/n_2}$
Paired t-test	Compare means of the same subjects under two conditions (before/after).	$t = \bar{d} / (s_d / \sqrt{n})$, where d = differences
Two-proportion z-test	Compare conversion / click-through rates in A/B tests.	$z = (\hat{p}_1 - \hat{p}_2) / \sqrt{\hat{p}(1 - \hat{p})(1/n_1 + 1/n_2)}$
Chi-squared (χ^2)	Test independence / goodness of fit for categorical data.	$\chi^2 = \Sigma (O - E)^2 / E$
ANOVA (F-test)	Compare means across 3+ groups (equal variances assumed).	F = between-group variance / within-group variance

6. Model Estimation - Regression & Classification

From Variance to Covariance to Correlation

Covariance: $\mathrm{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$ — generalizes variance to two variables.
Pearson correlation: $r = \mathrm{Cov}(X, Y) / (\sigma_X \cdot \sigma_Y) \in [-1, +1]$ — standardized and unit-free.
+1 = perfect positive linear | 0 = no linear relation | −1 = perfect negative linear.

Linear Regression (OLS)

Model: $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$, with ε_i i.i.d. and $E[\varepsilon] = 0$.
Ordinary Least Squares: minimize the sum of squared residuals

$$\min_\beta \ \Sigma_i (y_i - \beta_0 - \beta_1 x_i)^2$$

⇒ fitted: $\beta_1 = \mathrm{Cov}(X, Y) / \mathrm{Var}(X)$, $\beta_0 = \bar{y} - \beta_1 \bar{x}$
R² (coefficient of determination): $R^2 = 1 - SS_{res} / SS_{tot} \in [0, 1]$ — closer to 1 is better.

A good fit requires a linear relationship, residuals that are i.i.d. and approximately normal, and no strong multicollinearity. Watch for overfitting (fits noise) and underfitting (misses signal).

Logistic Regression (binary classification)

Models $P(Y = 1 \mid x)$ using the **sigmoid** of a linear combination of features:

$$P(Y = 1 \mid x) = \sigma(\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p)$$

where $\sigma(z) = 1 / (1 + e^{-z})$
Logit (inverse): $\log(p / (1 - p)) = \beta_0 + \Sigma \beta_j x_j$
Parameters are fit by **maximum likelihood** (solvers: L-BFGS, gradient descent). A train / test split is used to evaluate generalization.

Confusion Matrix & Classification Metrics

	Actual Positive	Actual Negative
Predicted Positive	TP (true positive)	FP (type I error)
Predicted Negative	FN (type II error)	TN (true negative)

Accuracy = (TP + TN) / (TP + TN + FP + FN) — overall correctness.
Precision = TP / (TP + FP) — of predicted positives, how many are correct?
Recall (Sensitivity) = TP / (TP + FN) — of actual positives, how many are found?
F1-score = 2 · (Precision · Recall) / (Precision + Recall) — harmonic mean, robust for imbalanced classes.

Accuracy can be misleading with imbalanced classes (e.g. predicting "sun" 95% of the time in July). Always check precision, recall and F1 alongside accuracy.